

address changes each time we connect to the Internet (unless we have a static IP connection).

But this assumption is false. Websites can still track you even though you are not willing to sign in or retain cookies. What they use to identify you is the *fingerprint of the browser* you use. Unlike a serial number, this is not something imposed on the browser by the developer. Instead, it is calculated by the websites based on the configuration and settings of the browser. They even consider the extensions that you use.

To put it simply, consider the example of identifying a vehicle. The most straightforward and accurate way would be to use the registration number (exclude spoofs). But in most cases, we simply say something like, "...a red SUV with an eagle sticker on the hood, tops removed, one mirror slightly broken." This would be more than enough since such a combination will be unique in a particular city.

Browsers can also be identified like this. The parameters start from basic configurations and the operating system to some unimaginable ones, such as:

- Extensions used
- Plugins and file formats supported
- Screen resolution and colour depth
- Language and time zone
- Touchscreen support

Quoting *panoptickclick.eff.org*, "Browser fingerprinting is a method of tracking Web browsers by the configuration and settings information they make visible to websites, rather than traditional tracking methods such as IP addresses and unique cookies."

How to test your browser and what to do next

We've seen how a website can track our browser using browser fingerprinting. Now let's figure out how susceptible our browser is. To check this, simply visit *panoptickclick.eff.org*, which is the result of an important research undertaken by EFF.

You can see a **TEST ME** button on the home page itself. Click on that, and your browser gets tested. It takes a while, and once complete, you are presented with a quick report showing how your browser is susceptible to different aspects of tracking. You can get a detailed report, and install the privacy extension *Privacy Badger* from the same page.

Blocking trackers with Privacy Badger

There are hidden trackers embedded in many websites in the form of scripts and cookies. They are used to track you, i.e., to create a record of the pages you visit, things that you like and things you don't. These details are usually used to serve personalised ads, and sometimes are sold to a third party for business purposes. The EFF Privacy Badger is a useful extension that blocks such trackers. Visit *eff.org/privacybadger* to install this add-on in your browser.

When Privacy Badger realises that the ads in a page are tracking you, it simply blocks the trackers from loading

Test	Result
Is your browser blocking tracking ads?	✗ no
Is your browser blocking invisible trackers?	✗ no
Does your browser unblock 3rd parties that promise to honor Do Not Track ?	✗ no
Does your browser protect from fingerprinting ?	✗ your browser has a unique fingerprint

Figure 1: *panoptickclick.eff.org* shows that the incognito mode of the Chromium Web browser is clearly susceptible to tracking

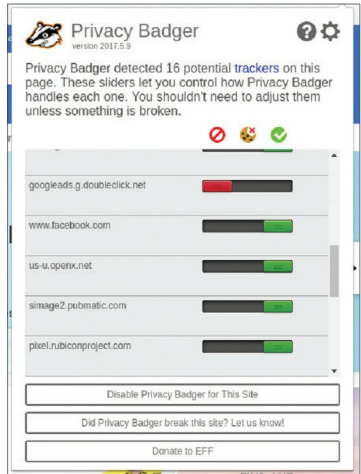


Figure 2: Privacy Badger detects 16 potential trackers on a popular shopping website

more content. To borrow the words used by EFF, "...to the advertiser, it's like you suddenly disappeared."

When you install Privacy Badger and visit a website with trackers, it will display a list of these with sliders if you click on its icon in the browser toolbar. The slider will probably be green the first time, which means that the tracker is still not tracking you and hence is allowed. As you continue browsing, the slider might turn yellow and then red, indicating Privacy Badger's opinion on them, and the blocking level.

You can say that Privacy Badger acts like an ad blocker and in fact, it is based on the code of AdBlockPlus. Still, it's a bit different. Its aim is not to block every ad, but to block every tracking element. This is important since not all ads